TANG JIAYI

AI Application Development | LLM & Agent Engineering | Natural Language Processing

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EDUCATION

National University of Singapore | M.Sc. in Artificial Intelligence for Science Aug 2026 – Jan 2028

Planned coursework: AIS5101 Applications of AI in Science; AIS5102 Practical Machine Learning for Scientific Discovery; AIS5103 Foundations of Deep Learning; AIS5104 Seminar Course for AI in Science; DSA4213 Natural Language Processing for Data Science; DSA5204 Deep Learning and Applications; DSA5207 Text Processing & Interpretation with Machine Learning.

Zhejiang University | B.Sc. in Pharmacy, GPA 3.65/4.00 Sep 2022 – Jul 2026

Relevant coursework: Calculus I/II, Linear Algebra, C Programming, Computational Biology for Drug Design, Biostatistics, and Medical AI.

PROJECT EXPERIENCE

MusicMol | Bidirectional Molecule–Music Generative AI Framework Co-first Author | 2025–2026

- Led project coordination and experiment integration, and contributed to music-to-molecule sequence modeling using REMI event representation, BPE tokenization, and an encoder–decoder Transformer to autoregressively decode MIDI sequences into SELFIES molecular strings.
- Processed 2 million ChEMBL compounds and supported inverse-model training on 10 million molecule–MIDI pairs; evaluated representations across 30 MoleculeACE tasks using Spearman correlation, R^2, RMSE, and local-neighborhood metrics.
- Contributed to an interactive AI system integrating molecular visualization, symbolic-music generation, real-time keyboard input, model inference, and candidate-molecule presentation; published as a ChemRxiv preprint (MusicMol, 2026)

Multi-label Protein Function Prediction with ESM Embeddings Project Lead | 2025

- Filtered 20,421 UniProt protein sequences by GO annotations and sequence length, producing a cleaned dataset of 6,407 sequences.
- Applied linear discriminant analysis (LDA) to compare CDT descriptors with ESM protein-language-model embeddings, confirming stronger class separability with ESM representations.
- Built XGBoost and MLP multi-label classifiers for five cellular-component categories, achieving an average F1 score of 0.70.

Natural-Language Multi-Track Arrangement Agent Personal Project | Design Stage

- Designed a multi-agent workflow that translates natural-language arrangement instructions into structured constraints for musical form, instrumentation, harmony, and texture, and coordinates specialised agents for planning, generation, validation, and revision.
- Planned LangGraph-based state orchestration, Pydantic-constrained structured outputs, and tool calling with music21 for MIDI/MusicXML generation and editing.
- Defined evaluation metrics for instruction adherence, structural validity, edit success, latency, reproducibility, and human preference.

TECHNICAL SKILLS

Programming & Frameworks: Python, SQL, Git, PyTorch, scikit-learn, Transformers, RDKit; **AI/ML:** Transformers, GNNs, embeddings, sequence generation, multi-label classification; **Agent Development (learning):** RAG, LangGraph, tool calling, structured outputs, FastAPI.

LEADERSHIP & HONORS

Vice President and Violinist, Zhejiang University Wenqin Symphony Orchestra | First Prize, 7th National College Student Art Exhibition | Zhejiang University Outstanding Student